MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
END SEMESTER EXAMINATIONS NOVEMBER- DECEMBER 2024

SOFT COMPUTING TECHNIQUES [ELE 4062]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 27 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Perform one epoch of training using Error back propagation algorithm for the data given below. The weight matrices for the 1st layer are given by $V = \begin{bmatrix} -0.5 & 1 \\ -1 & -1 \end{bmatrix}$, and $W = [-0.5 \quad -1 \quad -1]$, weight matrix V from input to hidden layer and, W hidden to output layer.

. The input to the network p=0.5 and bias input is -1. The desired output is $d = \sin(\pi p)$. Use unipolar continuous neurons for 1st layer and linear neuron for second layer. Take c=λ=1.

i) Calculate net^j, $f(net^j)$, net^o and f(net^o) $(i.e. output)$

ii)f'(net^j) and f'(net^o)

iii) Compute error signals δ_o and δ_y

iv)Draw the network with updated weights ***(05)***

1B. Compare two liquid level controllers using fuzzy operators mentioned below for their control levels and flow speed. The following values are available.

Flow Speed	Control Level 1	Control level 2
0	0	0
20	0.5	0.45
40	0.35	0.55
60	0.75	0.65
80	0.95	0.9
100	1	1

Given the universe of discourse as $X = \{0,20,40,60,80,100\}$, and the membership functions

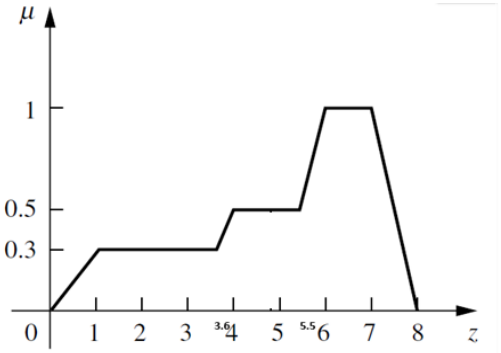
$$L_1 = \left\{\frac{0}{0} + \frac{0.5}{20} + \frac{0.35}{40} + \frac{0.75}{60} + \frac{0.95}{80} + \frac{1}{100}\right\}$$

$$L_2 = \left\{\frac{0}{0} + \frac{0.45}{20} + \frac{0.55}{40} + \frac{0.65}{60} + \frac{0.9}{80} + \frac{1}{100}\right\}$$

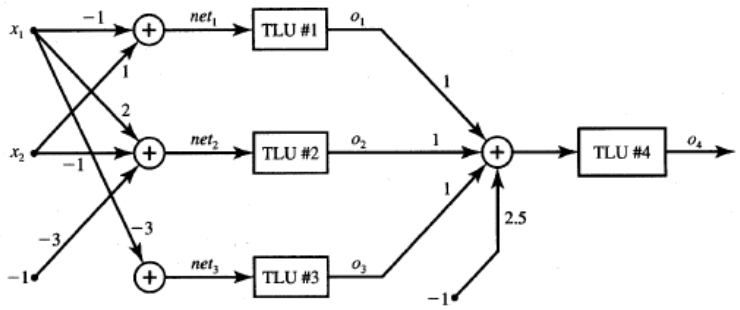
Determine the following.

i) $L_1 \cap L_2$ ii) $L_1 \cup L_2$ iii) $\overline{L_1 \cap L_2}$ ***(03)***

1C. Determine the defuzzified output using weighted average method, union of three membership functions are given below. ***(02)***



2A. Determine the segment of the x_1, x_2 plane for which $O_4 = 1$.
The feedforward network shown in Fig., using bipolar binary neurons is mapping the entire x_1, x_2 plane into a binary O value.



(04)

2B. Illustrate delta learning rule for one epoch of training using bipolar sigmoid activation function.

$$x_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix}; x_2 = \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}; x_3 = \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

Initial weight vector is

$$w^1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix}$$

The desired response for x_1, x_2, x_3 are $d_1 = -1$, $d_2 = -1$ and $d_3 = 1$; ***(04)***

2C. Determine the cartesian product $R = P \times Q$.

Two fuzzy sets are given as

$$P = \left\{\frac{0.1}{2} + \frac{0.3}{4} + \frac{0.7}{6} + \frac{0.4}{8} + \frac{0.2}{10}\right\}$$

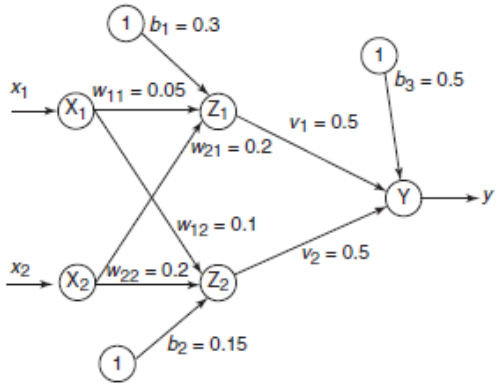
$$Q = \left\{\frac{0.1}{0.1} + \frac{0.3}{0.2} + \frac{0.3}{0.3} + \frac{0.4}{0.4} + \frac{0.5}{0.5} + \frac{0.2}{0.6}\right\}$$

(02)

3A. Implement XOR function using multiple adaptive linear neural network (MADALINE), with bipolar inputs (x1, x2) and targets(t). Initial weights and biases are given in the network in figure. Perform one epoch. Assume c = 0.5. ***(04)***

x_1	x_2	1	t
1	1	1	−1
1	−1	1	1
−1	1	1	1
−1	−1	1	−1

The activation function is bipolar binary.



3B. Determine the relationship between frequency and reliability index using i) max - min composition ii) max-product composition, for the given two relations.

$$R = \begin{array}{c} \\ 9 \\ 18 \\ 27 \\ 36 \end{array} \begin{array}{c} \begin{bmatrix} -100 & -50 & 0 & 50 & 100 \end{bmatrix} \\ \begin{bmatrix} 0.2 & 0.5 & 0.7 & 1 & 0.9 \\ 0.3 & 0.5 & 0.7 & 1 & 0.8 \\ 0.4 & 0.6 & 0.8 & 0.9 & 0.4 \\ 0.9 & 1 & 0.8 & 0.6 & 0.4 \end{bmatrix} \end{array}$$

$$S = \begin{array}{c} \\ -100 \\ -50 \\ 0 \\ 50 \\ 100 \end{array} \begin{array}{c} \begin{array}{ccccc} 2 & 4 & 8 & 16 & 20 \end{array} \\ \begin{bmatrix} 1 & 0.8 & 0.6 & 0.3 & 0.1 \\ 0.7 & 1 & 0.7 & 0.5 & 0.4 \\ 0.5 & 0.6 & 1 & 0.8 & 0.8 \\ 0.3 & 0.4 & 0.6 & 1 & 0.9 \\ 0.9 & 0.3 & 0.50 & 0.7 & 1 \end{bmatrix} \end{array}$$

R is the relationship between frequency and temperature, S is the relationship between temperature and reliability index of a circuit. ***(04)***

3C. Determine i)$B_{0.6}$ ii) $(A \cup B)_{0.5}$
For the fuzzy sets A and B.

$$A = \left\{\frac{1}{1} + \frac{0.8}{2} + \frac{0.65}{3} + \frac{0.4}{4} + \frac{0.12}{5}\right\}$$

$$B = \left\{\frac{0.2}{1} + \frac{0.5}{2} + \frac{0.67}{3} + \frac{0.85}{4} + \frac{1}{5}\right\}$$ ***(02)***

4A Design a temperature-humidity-based fuzzy logic control (FLC) system for an air conditioning (AC) system which adjusts the AC settings based on both temperature and humidity levels. Use 3 membership for the input and output variables. Create a set of rules for controller action and get the defuzzified values. The descriptors are

- Temperature: Low (20–24°C), Medium (24–28°C), High (28–32°C)
- Humidity: Dry (0–30%), Comfortable (30–60%), Humid (60–100%)

(05)

Output

- AC power Level: Low (0-30%), Medium (30-60%), High (60-100%)

Example Rules:

1. If temperature is High and humidity is Humid, then AC Power Level is High.
2. If temperature is Medium and humidity is Comfortable, then AC Power Level is Medium.
3. If temperature is Low and humidity is Dry, then AC Power Level is Low.

Use triangular membership for input and output. Determine the crisp output when the temperature is 26°C and Humidity 50.

4B. Explain Fuzzy inference system with a neat block diagram ***(03)***

4C. Draw the flow chart of Hebb training algorithm ***(02)***

5A. Draw the flow chart of neuro fuzzy system operation. ***(03)***

5B. Determine the value of X corresponding to maximum value of the given function $y=0.5x^{\wedge 3}$,where X is an integer in the range [0,31],using Genetic Algorithm. use 5-bit binary string. The initial population: [10001], [11110], [10101], [01111]. Perform single point cross-over after 2nd bit. The mutation chromosome is [01000]. Determine the average fitness after one iteration. ***(05)***

5C. How genetic algorithms can improve the neural network model? ***(02)***